Development of continuous, protein-specific predictors of the impact of protein sequence variants

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BACKGROUND

Up to date, computational models developed for predicting the effect of protein sequence variants have mostly focused on predicting the binary effect (benign/pathogenic)[1]. New research efforts are made in producing continuous estimates comparable to functional assays, as they contribute to the mechanistic interpretation of the impact of variants[2]. Here, we present our approach to this problem following the protein-specific method. We collected deep mutational scanning experiments data available in the literature[1,3-6] for building regression models for individual proteins. We trained protein specific predictors (PSP) for each protein in our dataset with a set of sequence- and structure (AlphaFold[7])-based input features to predict the functional impact of variants. We also found that our PSPs give reasonable results when applied to her proteins, opening the possibility of generalization of our method to the proteins in the clinical genome.

OBJECTIVES

1. Predicting continuous functional changes upon mutations with protein-specific approach, by building regression models for 29 proteins separately

2. Using protein-specific tools for predicting unknown variants in other proteins for testing the power of PSP on other proteins

METHODOLOGY

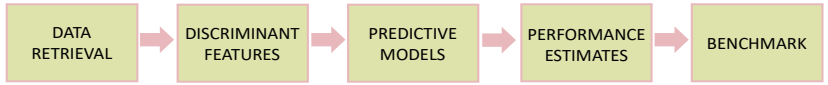

Figure 1. *Workflow for building a predictor*

DATA RETRIEVAL: DMS studies are collected for 29 unique proteins from literature[1,3-6]. The assays were then normalized following the Gray et al.[1] normalization scheme.

DISCRIMINANT FEATURES: 10 features are used for the models that can be grouped into two: **Conservation/sequence-based features**, calculated through use of amino-acid replacement or multiple sequence alignment information; and **Structural features**, derived from AlphaFold[7] structure predictions.

PREDICTIVE MODELS: To handle binary distribution seen in datasets, after each train/test split, training sets are fit to Gaussian mixture models[8] and most populated groups are undersampled. Then, Multiple Linear Regression and Random Forest Regression models are built for each protein[8].

PERFORMANCE ESTIMATES: A stringent leave-one-position-out (LOPO) method is applied for estimating the performance of models where on each iteration, *variants at the same position* are separated for testing.

BENCHMARK: LOPO performances are compared to Envision[1], a predictor of similar nature. Further, we compared our models to 19 available in silico tools. Lastly, each protein-specific model was used for predicting variant in other proteins in our dataset for cross-prediction.

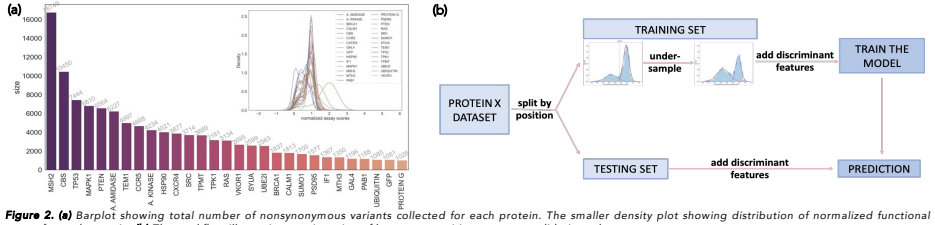

Figure 2. ***(a)*** *Barplot showing total number of nonsynonymous variants collected for each protein. The smaller density plot showing distribution of normalized functional assays for each protein.* ***(b)*** *The workflow illustrating one iteration of leave-one-position-out cross-validation scheme.*

RESULTS

Performance of 29 unique Protein-Specific Predictors

The results of the PSPs show that, the method works moderately well for most of the proteins, e.g. TEM1, TP53, BRCA1 (Figure 3); whereas lower for others (e.g. CALM1 with spearman rank 0.2, SYUA with spearman rank 0.22 etc.). Overall, the spearman's rank correlations varied between 0.2 and 0.68.

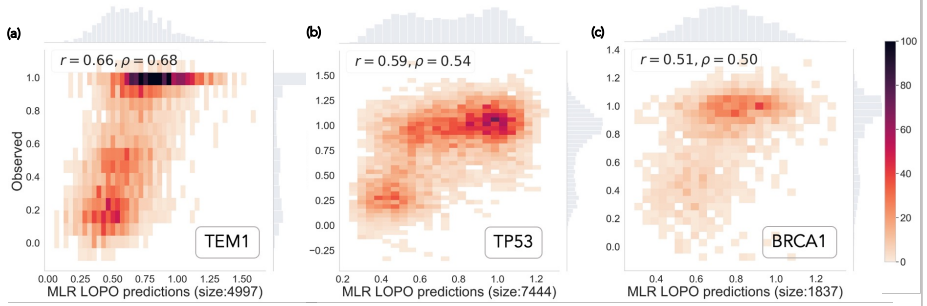

Figure 3. *Heatmaps showing the predictive performance of MLR models for three proteins;* ***(a)*** *TEM1,* ***(b)*** *TP53 and* ***(c)*** *BRCA1. Y-axes show normalized functional assay values for each protein, and x-axes are the predictions obtained from our models. Pearson and spearman rank values are shown on upper left.*

Comparison of our in-house predictors to other available in silico tools

We did comparisons of MLRs against ENVISION, a predictor which also uses functional assays for training. We observed that both approaches outperform each other, in accordance with our previous results for the clinical phenotype prediction[9]. Further, we compared our MLR and RFR models against around 20 prediction tools[10], and observed that our models are performing on the average range of those binary predictors.

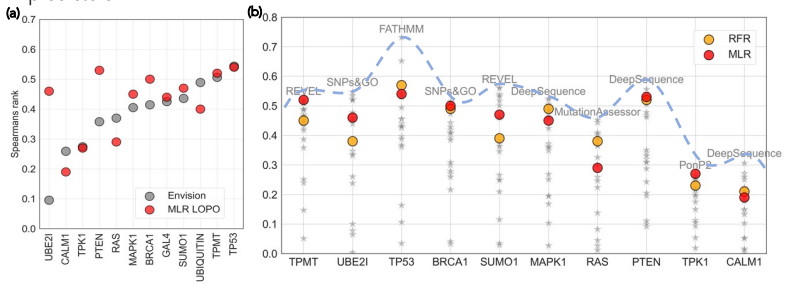

Figure 4. *Comparison of the performance of our in-house PSPs.* ***(a)*** *Fair comparisons between MLR and Envision, which are both predictors trained using deep mutational scanning experiments.* ***(b)*** *Comparison of MLR and RFR models against available in silico tools. Proteins on x-axis are sorted by the median of spearman ranks of 19 tools, and dashed blue line show the highest values.*

Some specific models work for other proteins: clearing the way for application of PSPs to clinical genome

We used PSPs for predicting variants in other proteins. We used PSPs that gave the highest correlation and compared against Envision. The same trend of outperforming Envision was observed again, supporting the hypothesis according to which a given PSP can serve for several proteins.

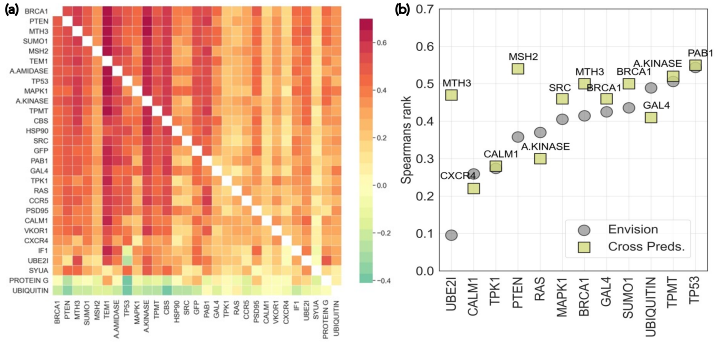

Figure 5. *Performance of cross-predictors.* ***(a)*** *Heatmap showing spearman's rank correlation of cross-predictors.* ***(b)*** *Comparison PSPs that give the highest correlation for the predicted protein against Envision*

CONCLUSION

In summary, we developed protein-specific models for predicting the functional assay values. Overall, our results indicate that the models provide moderately good predictions of functional assays. Comparisons toexisting tools showed that, our choice of features were good enough for obtaining compatitive predictors.Our results with cross-predictions indicate a possible way to advance in the development of PSPs by creating a mapping between the predictor and the protein spaces.

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